Understanding the IDSP outbreak dataset

A guide to the source data, our standardisation work and responsible use

What is IDSP

The Integrated Disease Surveillance Programme (IDSP) is India's programme for detecting and responding to epidemic prone diseases. Launched in November 2004, it operates under the Ministry of Health and Family Welfare. The National Centre for Disease Control (NCDC) publishes weekly outbreak bulletins based on reports received from states and union territories. [1, 2]

An outbreak is the occurrence of more cases of a disease than expected in a particular place and period. The bulletins describe reported outbreak events, their locations, cases, deaths and response measures. Our dataset brings these scattered PDF tables into a structured historical series that can be filtered and analysed.

The wider IDSP system collects suspected, presumptive and laboratory confirmed cases through S, P and L reporting forms. The dataset described here is compiled from weekly outbreak bulletins; it is not the full S, P and L reporting database. A case in an outbreak bulletin should not automatically be treated as laboratory confirmed. [1, 4]

What data is available

Dimension	What the dataset covers
Coverage	Weekly outbreak reports from 2009 onwards. The supplied quality review includes records through 2026; this does not establish complete coverage of 2026. [4, 5]
Geography	States and union territories, with district information where reported. Some entries cover all districts or do not identify a usable district.
Disease and impact	Reported disease or illness, associated ICD codes where mapped, and reported numbers of cases and deaths.
Time	Bulletin year and week, outbreak start date and reporting date where available; the reference datasheet also describes month and date based access.
Context	Bulletin section, outbreak identifier where available, current status, source comments and cleaning notes, as described in the quality review.

What one row means

One row represents an outbreak report in a bulletin, not an individual patient and not necessarily a unique outbreak. Follow-up reports may describe an event already reported, while an event affecting more than one district may have separate rows. Row totals therefore measure reports unless records have been reconciled into distinct events.

How to read the reporting sections

Section	Meaning	How to use it
New alerts	Outbreaks listed in the bulletin's alert section.	Use the source section label; it does not by itself prove reporting within seven days of onset.
Reported late	Outbreaks listed as reported late.	The bulletin week and outbreak start period may differ. Reporting dates are often absent from this section's layout.
Follow-ups	Updates on outbreaks reported earlier.	Do not add these reports to earlier records without checking whether counts overlap or revise earlier figures.

How we make the source usable

We organise the bulletin tables into consistent fields, check the identity of each bulletin, standardise names and date values, and retain notes where source values are changed, blanked or interpreted. Standardisation makes records easier to compare; it does not fill gaps in surveillance or resolve every source ambiguity. [4, 5]

Work undertaken	Example
Check the actual bulletin	A link labelled week 44 of 2010 serves week 44 of 2013. The review identifies 11 mislabelled links, including 10 duplicate copies. Checking the PDF's own identity avoids assigning it to the wrong period or ingesting the same bulletin twice.
Reconstruct irregular tables	Merged state cells, changing headers and names split across lines can displace or lose values during extraction. The review records more than 30 header variants and a nested table of four chickenpox outbreaks in Madhubani in 2018 week 8.
Standardise geography	The reference datasheet identifies the Local Government Directory as the source for state and district names. Ambiguous names and district aggregates need separate treatment; a standard name does not establish a common historical boundary.
Harmonise disease names	Spelling aliases, line breaks and variants such as "(P. falciparum)" and "(P.falciparu m)" are reconciled to support consistent grouping.

Work undertaken	Example
Retain an audit trail	The notes field preserves source text when a value is changed, blanked or interpreted.

What the added ICD codes mean

We add codes from the World Health Organization's International Classification of Diseases (ICD) to help users group disease names consistently. These codes are an enrichment to the outbreak data, not a new diagnosis or a confirmation of the source diagnosis. [3, 5]

For a combined entry such as "Dengue & Scrub Typhus" or "Hepatitis A & E", the relevant codes are separated by " | ". The reported cases remain a shared total; no disease wise split is invented. The review identifies 635 reports naming multiple diseases.

We leave the ICD field blank where no suitable mapping has been assigned, including some locally named conditions or source labels that cannot be mapped confidently. The review lists "Others", Chandipura virus variants and "Amoebic Encephalitis" among unmapped entries. A blank means unmapped in this dataset, not that WHO necessarily lacks a code.

Common source errors and how we handle them

The examples below come from the supplied quality review. They illustrate both repairs and cases where uncertainty remains visible. [5]

Source issue	Example	Treatment and implication
Malformed date	12.056.09 in 2009 week 25	Published as 12-06-2009 after checking the source context.
Wrong year in a date	01-08-19 in a Karnataka row in 2015 week 31	Published as 01-08-2015. A date can be syntactically valid but inconsistent with the bulletin context.
Period given instead of day	"May 2010" or "1st wk of August"	Represented by the midpoint of the stated period, with the interpretation recorded in notes.
Damaged ID punctuation	MP/SGR//2016/1/0025	Unambiguous separator errors are repaired. IDs damaged beyond reliable repair are blanked, with the original text retained in notes.
Placeholder instead of value	"???" or a string of dashes	Published as null, with source text retained in notes. Null is not zero.
Value in the wrong column	Dates printed in the Current Status box in 2010 week 23	Such issues have been rectified by placing the information in appropriate columns
Uncertain date order	Reporting date earlier than start date	Both dates are retained as printed when the source does not establish which is wrong.
Missing or non-unique IDs	A follow-up pointer in place of an ID, or one ID used on several rows	Cross-reference text is retained in notes. IDs begin in 2016; earlier blanks are expected.

Missing and unclear information

Some weekly reports are unavailable or do not contain outbreak details. Missing information has not been estimated or treated as zero. The absence of a report does not necessarily mean that no outbreaks occurred during that week.

Where errors or inconsistencies could not be resolved confidently, we have retained the available information and documented the limitation rather than making assumptions. Users should consider these gaps when comparing outbreak patterns across places or periods.

What to check before analysis

Measure reported outbreaks

This is a record of outbreaks received and published through the surveillance system, not a complete count of all illnesses or deaths in India. Differences between places or years can reflect reporting coverage as well as disease occurrence.

Choose the time measure

Use bulletin year and week for publication patterns and outbreak start dates for onset patterns. Reporting dates describe another stage. Follow the bulletin's stated week rather than silently replacing it with another calendar convention. Exclude imputed period dates from analyses requiring exact days.

Check repeated and combined reports

Separate alerts, late reports and follow-ups according to your question. Do not deduplicate on unique_id alone. Do not copy a multi-disease case total into each disease category and then sum it.

Check geography before adding totals

Historical changes involving Jammu and Kashmir and Ladakh, or the merged Dadra and Nagar Haveli and Daman and Diu territory, require a deliberate boundary choice.

Distinguish missing values from zero

A blank may reflect missing source information, a column absent from a section, an unusable value or an unmapped code.

Check what the numbers count

A 2021 week 8 avian influenza entry reports “03 birds” and “425 birds” in the case and death fields. Exclude animal counts from human totals. “Several thousand fever cases” is published as null rather than a guessed number.

Sources and documentation

[1] IDSP About IDSP. Programme background and S, P and L reporting forms. https://www.idsp.mohfw.gov.in/index1.php?lang=1&level=0&lid=1592&linkid=313

[2] IDSP Weekly Outbreaks. Official bulletin archive. https://www.idsp.mohfw.gov.in/index4.php?lang=1&level=0&lid=3689&linkid=406

[3] WHO International Classification of Diseases. https://www.who.int/standards/classifications/classification-of-diseases

[4] DataSheet_IDSP.pdf. Supplied reference datasheet; source for the dataset description and Local Government Directory standardisation reference.

[5] IDSP_Outbreak_Bulletins_Explainer.docx. Supplied internal quality review; source for processing examples, issue counts and remaining limitations. Counts refer to that review’s snapshot, not a fresh audit of the underlying dataset.